# Garrett W. Thrash

M.D. Candidate (c/o 2027) • Incoming Ph.D. Student, Neuroengineering (2027)
University of Alabama at Birmingham Heersink School of Medicine
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Citizenship: United States  Languages: English

## Education

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|---|---|
| 2027 – 2032 (exp.) | **Ph.D., Neuroengineering**<br>University of Alabama at Birmingham, Birmingham, AL |
| 2023 – 2027 (exp.) | **M.D., Distinction in Health Equity**<br>UAB Heersink School of Medicine, Birmingham, AL |
| 2020 – 2023 | **B.S., Chemistry, *magna cum laude***<br>Birmingham-Southern College, Birmingham, AL |

## Licensure and Certification

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| 2024 – present | Nationally Registered Emergency Medical Technician (NREMT) |
| 2024 – present | Advanced Life Support (ALS) |
| 2024 | FEMA IS-100.c, IS-230.e, IS-700.b Emergency Management |
| 2023 – present | Basic Life Support (BLS) |

## Board Scores

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| USMLE Step 1 | Pass |
| USMLE Step 2 CK | 263 |

## Honors and Awards

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| 2026 – present | Alpha Omega Alpha (AOA) Honor Medical Society |
| 2025 – 2026 | Clerkship Honors: Internal Medicine, Neurology, Pediatrics, Family Medicine |
| 2025 – present | Gold Humanism Honor Society (GHHS) |
| 2024 – present | Health Equity Scholar, UAB Heersink School of Medicine |
| 2023 – present | Comer Memorial Scholarship, UAB Heersink School of Medicine |
| 2023 | Clyde Stanton Outstanding Senior in Chemistry Award, Birmingham-Southern College |
| 2022 – present | Mortar Board Honor Society |
| 2022 – 2023 | American Chemical Society |
| 2022 – 2023 | Alpha Epsilon Delta Honor Society |
| 2022 – 2023 | Kappa Mu Epsilon Honor Society |
| 2022 – 2023 | Alpha Lambda Delta Honor Society |
| 2020 – 2023 | Birmingham-Southern Presidential Scholarship |

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| 2020 – 2023 | Birmingham-Southern Football SAA Honor Roll |
| 2020 – 2023 | ALFA Insurance Scholarship |
| 2020 – 2023 | Shirley Dalton Memorial Medical Scholarship |

## Professional Memberships and Societies

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| 2026 – present | Student Member, Alabama Academy of Neurology |
| 2025 – present | Society of NeuroInterventional Surgery |
| 2023 – present | Medical Student Committee, Congress of Neurological Surgeons |
| 2023 – present | Student Member, American Association of Neurological Surgeons |
| 2023 – present | Student Member, North American Neuromodulation Society |
| 2020 – present | Sigma Alpha Epsilon Fraternity |

## Peer Review Activities

None to date.

## Major Research Interests

Deep brain stimulation, neuromodulation, and brain–computer interfaces; neuroengineering and neuroprosthetics; global and equitable access to neurological care.

## Research Experience

2026 – present **Walker Research Lab, UAB Neurology**
Dr. Harrison Walker
- Machine-learning decoding of subthalamic nucleus recordings for adaptive deep brain stimulation and brain–computer interfaces; planned dissertation laboratory for the Neuroengineering Ph.D.

2024 – present **Global Neurosurgery Research Group, Mission:Brain Foundation**
Coordinator
- Coordinate roughly 35 international medical graduates, mainly from low- and middle-income countries, in a multi-project research group advancing equitable access to neurological care.
- Lead members across many time zones and backgrounds toward shared goals; sharpen scientific writing, study design, and critical review across the group.

2023 – present **Holland Clinical Research Lab, UAB Neurosurgery**
Dr. Marshall Holland
- Joined as a first-year medical student on a single systematic review; grew into co-leading lab meetings and mentoring younger medical students and undergraduates.
- Chart review, data collection, study design, manuscript writing and review; managed submission and revision for multiple manuscripts and IRB protocols.
- Designed independent studies and authored original IRB protocols.

2022 – 2023 **Immunology Research, UAB**
Dr. Chander Raman
- Investigated ultraviolet B–induced immune suppression in murine models and cell lines, including vitamin D3 effects on samples from multiple sclerosis and lupus patients.
- Cell culture, flow cytometry, and confocal microscopy; cryosectioning and staining of lymph node, spleen, and skin tissue.
- Oversaw breeding, treatment administration, and upkeep of multiple transgenic mouse lines.
- High-dimensional flow cytometry analysis in FlowJo (tSNE, UMAP, TriMap).

2021 – 2022 **Enzyme Kinetics Research, Birmingham-Southern College**
Dr. Kate Hayden
- Expressed novel enzymes from vector-transfected plasmids; purified target proteins by SDS-PAGE, centrifugal size exclusion, and His-tag affinity chromatography.
- Kinetic assays on novel enzymes (tvGAPDH, cdGAPDH, cpPK); Michaelis–Menten and Lineweaver–Burk modelling of $K_m$ and $V_{max}$ for medicinal-chemistry collaborators.

## Grants and Fellowships

07/2026 – 07/2027 Doximity Op-Med Medical Writing Fellowship
Selected through a competitive national application open to attendings, residents, and medical students for one of 40 fellowship positions. Write narrative essays and op-eds on a range of medical topics for Op-Med, Doximity's physician-facing publication.

## Peer-Reviewed Publications

Bold indicates the CV owner. Listed newest first.

[1] Sun Y, Ramapuram H, Barrentine B, **Thrash GW**, et al. Neighborhood Socioeconomic Status, Insurance, and Race Affect Presentation and Outcomes Following Endoscopic Pituitary Adenoma Resection. *Acta Neurochirurgica* (Oct. 1, 2026). doi:10.1007/s00701-026-07037-5.

[2] **Thrash GW**, Stacey GK, Wang E, et al. Therapeutic Deep Brain Stimulation for Restless Legs Syndrome: A Systematic Review and Meta-Analysis. *Neuromodulation: Technology at the Neural Interface* (Sept. 2026). doi:10.1016/j.neurom.2026.08.008.

[3] **Thrash GW**, DeMessie B, Wang EL, et al. Multi-Pedicle versus Single-Pedicle Middle Meningeal Artery Embolization for Nonacute Subdural Hematoma. *Neuroradiology* (Sept. 21, 2026). doi:10.1007/s00234-026-04187-2. PMID: 42766032.

[4] Toussaint F, dos Santos Rubio EJ, Agwu C, … **Thrash GW**, et al. A Scoping Review of Non-Governmental Organizations in Global Neurosurgery: Contributions, Gaps, and Future Directions. *Acta Neurochirurgica* (Aug. 26, 2026). doi:10.1007/s00701-026-07014-y.

[5] Wang EL, **Thrash GW**, Razi R, et al. Spinal Cord Stimulation Effects on Restless Legs Syndrome: A Single Center Experience. *Stereotactic and Functional Neurosurgery* (Aug. 7, 2026), 1–19. doi:10.1159/000553453. PMID: 42566390.

[6] **Thrash GW**, Wang E, Brockington D, et al. Neuromodulation for Restless Legs Syndrome: A Systematic Review and Mechanistic Considerations for Spinal Cord Stimulation. *Sleep and Breathing* 30.1 (Mar. 2026). doi:10.1007/s11325-026-03638-7. PMID: 41806217.

[7] Feldman MJ, Hereford WJ, Gaskin R, **Thrash GW**, et al. Fortify Alabama: Corn Masa Fortification for Spina Bifida Prevention. *Pediatrics* 157.2 (Feb. 2026). doi:10.1542/peds.2025-073306. PMID: 41570999.

[8] Dos Santos Rubio E, Brice KS, Toussaint F, … **Thrash GW**, et al. Governance and Policy in Global Neurosurgery: A Scoping Review of National and International Efforts. *Neurosurgical Review* 48.1 (Oct. 2025), 750. doi:10.1007/s10143-025-03914-2. PMID: 41171440.

[9] Holland MT, Mekhail MN, **Thrash G**, et al. Adaptive Spinal Cord Stimulation Improves Restless Legs Syndrome: Case Report, Literature Review, and Mechanistic Hypothesis. *Sleep Medicine* 134 (July 2025), 106664. doi:10.1016/j.sleep.2025.106664. PMID: 40652630.

[10] Kamath A, **Thrash GW**, Rocque BG, et al. Psychological Functioning in Children with Hydrocephalus: A Scoping Review. *Journal of Neurosurgery: Pediatrics* 36.4 (July 2025), 404–420. doi:10.3171/2025.3.PEDS24533. PMID: 40680304.

[11] **Thrash GW**, Evans RE, Sun Y, et al. Stereotactic Radiosurgery Treatment of Pediatric Arteriovenous Malformations: A PRISMA Systematic Review and Meta-Analysis. *Child's Nervous System* 41.1 (May 2025), 188. doi:10.1007/s00381-025-06835-z. PMID: 40407914.

[12] Sun Y, Laskay NMB, **Thrash GW**, et al. The Association of Area Deprivation Index and Spine Surgery Outcomes: A Systematic and Narrative Review. *Operative Neurosurgery* 29.6 (Apr. 2025), 775–784. doi:10.1227/ons.0000000000001551. PMID: 40168531.

[13] **Thrash GW**, Wang E, Sun Y, et al. Clinical Trials in Neuromodulatory Treatment of Drug-Resistant Hypertension and the Need for Spinal Cord Stimulation Trials: A PRISMA Systematic Review. *Bioelectronic Medicine* 10.1 (Dec. 2024), 28. doi:10.1186/s42234-024-00160-7. PMID: 39617923.

[14] **Thrash GW**, Hale AT, Feldman MJ, et al. Pediatric Non-Galenic Pial Arteriovenous Fistula's Characteristics and Outcomes: A Systematic Review. *Child's Nervous System* 40.6 (Mar. 2024), 1721–1729. doi:10.1007/s00381-024-06352-5. PMID: 38506930.

***In Press***

1. Thrash GW, Evans RE, Rocque B, Blount JP, Johnston J, Rozzelle C, Jones J. Surgical versus endovascular treatment of pediatric intracranial aneurysms: a systematic review, meta-analysis, and meta-regression. *Accepted, pending publication.*

***Submitted / Under Review***

1. Thrash GW, Stacey GK, Razi R, Rives C, Massey H, Brockington D, Cho H, Boudreau H, George J, Holland MT. Diverging perceptions between program leadership and applicants on the neurosurgery arms race to research: a single-center survey. *Neurosurgical Review*.
2. Thrash GW, Stacey GK, Wang E, Massey H, Jones JG. Coronary artery disease as a predictor for resolution of nonacute subdural hematoma after middle meningeal artery embolization: a single-center study. *Interventional Neuroradiology*.
3. Del Campos I, Thrash GW, …, Still M. Early predictors of poor outcomes in traumatic brain injury in a resource-limited setting: a retrospective study from Sierra Leone. *World Neurosurgery*.
4. Azan A, Thrash GW, …, Still M. Global suturing mission: the role of global surgical workshops for equitable access to surgical education. *Journal of Surgical Education*.
5. Coracan J, Azan A, Thrash GW, …, Quiñones-Hinojosa A. International hackathon to build an equitable innovation landscape in global neurosurgery. *Neurosurgery*.

## Editorials and Non-Peer-Reviewed Publications

Narrative essays and op-eds in preparation as a Doximity Op-Med Medical Writing Fellow (2026–2027); first pieces forthcoming.

## Books and Book Chapters

None to date.

## Patents and Intellectual Property

None to date.

## Oral Presentations

1. **Thrash GW**, Holland MT. Restless legs syndrome and spinal cord stimulation. UAB Neurosurgery Teaching Conference, June 2026.
2. **Thrash GW**, Boudreau H*, Stacey GK, Razi R, Rives C, Massey H, Brockington D, Cho H, George J, Holland MT. Diverging perceptions between program leadership and applicants on the neurosurgery arms race to research: a single-center survey. Galbraith Neurosurgery Research Symposium, Birmingham, AL, 2025. *Presented by second author.
3. **Thrash GW**. UAB Mission:Brain chapter pledge. Boston Declaration of Global Neurosurgery, Kennedy Center, Boston, MA, April 24, 2025.
4. **Thrash GW**, Sun Y, Del Bene V, Bentley N. Intracranial recordings toward brain mapping of executive function: a systematic review. NeuroGateways Consortium of Neuroengineering and Brain-Computer Interfaces, Birmingham, AL, Spring 2024.
5. **Thrash GW**, Mier-Aguilar C, Xu H, Raman C. UVB induction of senescent cells. Senior thesis, Birmingham-Southern College Department of Chemistry, Birmingham, AL, May 2023.

## Abstracts and Poster Presentations

1. **Thrash GW**, Stacey GK, Wang E, Massie H, Maleknia P, Jones JG. Coronary artery disease, illicit drug use, and age as predictors of radiographic resolution of chronic subdural hematoma after middle meningeal artery embolization: a single-center experience. Society of NeuroInterventional Surgery (SNIS) Annual Meeting, Seattle, WA, July 2026.
2. **Thrash GW**, Wang E, Essibayi MA, Jones JG. Multi-pedicle versus single-pedicle middle meningeal artery embolization for nonacute subdural hematoma. Society of NeuroInterventional Surgery (SNIS) Annual Meeting, Seattle, WA, July 2026.
3. Wang E*, **Thrash GW**, Sun Y, Holland MT. Spinal cord stimulation effects on restless legs syndrome: a single-center experience. Annual Meeting, Congress of Neurological Surgeons, Los Angeles, CA, October 2025. *Presented by first author.
4. Uppalappati SC*, **Thrash GW**, Bowers C, Holland MT. Subdural empyema and concomitant SARS-CoV-2 infection: a PRISMA systematic review. Annual Meeting, Congress of Neurological Surgeons, Los Angeles, CA, October 2025. *Presented by first author.
5. **Thrash GW**, Wang E, Roberts AC, Sun Y, Walker HC, Shirvalkar P, Holland MT. SCS of RLS: a single-center retrospective survey. Neural Interfaces, Washington, D.C., June 2025.
6. **Thrash GW**, Wang E, Roberts AC, Sun Y, Walker HC, Shirvalkar P, Holland MT. Neuromodulation of restless legs syndrome. Neural Interfaces, Washington, D.C., June 2025.
7. **Thrash GW**, Wang E, Sun Y, Walker HC, Shirvalkar P, Becker BK, Holland MT. Neuromodulation of the autonomic nervous system: a systematic review. NeuroGateways Consortium of Neuroengineering and Brain-Computer Interfaces, Birmingham, AL, Spring 2024.

## Invited Lectures

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| July 2024 | Invited Speaker, Rural Health. UAB Heersink School of Medicine Social Determinants of Health Grand Rounds. |
| 2024 | Invited Speaker, Systematic Review Methodology. Project:RISE Research Equity Lecture. |

## Software and Projects

2026 **AI-assisted title and abstract screening pipeline**

Hybrid human–AI screening for a scoping review of roughly 10,000 records

- A language model votes on every record with a confidence and an exclusion code. High-confidence exclusions are validated against blinded human screening of a random sample with a pre-declared error threshold before they are trusted; the model then serves as second reviewer on the retained set, with human adjudication of every disagreement.
- Designed to meet the 2025 Cochrane/Campbell/JBI/CEE position statement and the RAISE recommendations on AI in evidence synthesis. Python, Anthropic API, Covidence.

## Technical Skills

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| Programming | Python (NumPy, pandas, SciPy, Matplotlib, scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, DeepLabCut), MATLAB, R |
| Data analysis | Neural signal processing of intracranial local field potentials and ECoG; high-dimensional flow cytometry analysis in FlowJo (tSNE, UMAP, TriMap); enzyme kinetics modelling (Michaelis–Menten, Lineweaver–Burk) |
| Clinical research | Systematic review and meta-analysis (PRISMA, meta-regression); study design and IRB protocol authorship; retrospective chart review and data abstraction from the electronic health record; survey design; patient-reported outcome instruments (IRLSS, NIH PROMIS); manuscript submission and peer-review response |
| Wet laboratory | Mammalian cell culture; flow cytometry and confocal microscopy; cryosectioning and immunostaining; tissue harvest and enzymatic digestion (lymph node, spleen, skin); transgenic mouse colony management and treatment administration; recombinant protein expression from vector-transfected plasmids; protein purification (SDS-PAGE, centrifugal size exclusion, His-tag affinity chromatography); enzyme kinetic assays |
| Surgical | First assist on cranial neurosurgical cases during the neurosurgery acting internship: sterile technique, gowning and gloving; patient positioning and Mayfield pinning; burr hole and craniotomy assistance (drilling under supervision, bone flap handling); microsurgical assistance under the operating microscope (suction and irrigation, bipolar hemostasis); dural, galeal, and skin closure (interrupted, running, and subcuticular suturing, knot tying, staples) |
| Clinical | Inpatient neurosurgery and neurology: serial neurological examinations with GCS and NIHSS scoring; acute stroke and neurotrauma consults; interpretation of CT, MRI, and catheter angiography; external ventricular drain management and intracranial pressure interpretation; subgaleal and Jackson–Pratt drain removal, suture and staple removal, and wound care; inpatient neurology consult and outpatient neurology clinic rotations; EHR documentation (Epic); NREMT-certified EMT with Advanced and Basic Life Support and prehospital emergency care in high-volume event settings |
| Tools | Jupyter, VS Code, Google Colab, Git and GitHub, LaTeX, Zotero, Covidence, REDCap, Box |

## Additional Training and Professional Development

2026 – present **Neuromatch Academy, Computational Neuroscience curriculum**
Self-paced study of the open course materials
- Working through the full NMA computational neuroscience sequence: Python for neural data, model fitting and comparison, generalized linear models, dimensionality reduction, and EEG/MEG analysis.

## Teaching and Mentoring

2024 – 2025 **Multidisciplinary Research Stewardship, UAB**
Neurosurgery Chair
- Hosted office hours and presentations on research fundamentals for medical students; connected students interested in neurosurgery with career mentors.

2021 – 2023 **Academic Resource Center, Birmingham-Southern College**
Chemistry Tutor
- Tutored undergraduate peers in analytical and organic chemistry and biochemistry.

2021 **Cell and Molecular Biology, Birmingham-Southern College**
Teaching Assistant
- Led the cell and molecular biology laboratory sections, including all grading and organizational tasks.

***Mentees: Medical Students***

| Mentee | Institution | Role / Focus | Dates | Outcomes |
|---|---|---|---|---|
| Grayson Stacey | UAB Heersink | Career guidance and research mentorship; co-author on scientific writing | Jan 2025 – present | 1 publication, 2 manuscripts under review, 1 abstract, 1 oral presentation |
| Riyaz Razi | UAB Heersink | Research mentorship, co-author on scientific writing, residency mentorship | Aug 2024 – present | 2 publications, 1 manuscript under review, 1 oral presentation |

***Mentees: Undergraduate Students***

| Mentee | Institution | Role / Focus | Dates | Outcomes |
|---|---|---|---|---|
| Rohma Syed | UAB | Mentorship on medical careers and medical school applications | 2025 – present | Medical school applications, Mission:Brain, research |
| Elijah Wang | UAB | Mentorship on medical careers and applications; co-author and first author on manuscripts | Sep 2023 – present | 5 publications (1 as first author), 1 manuscript under review, 6 abstracts (1 as first author) |

## Leadership and Committee Service

**Mission:Brain — UAB Chapter President; Regional Chair of Outreach; Coordinator and Lead Advisor, Global Neurosurgery Research Group**

- Led the UAB chapter: launched Project:RISE (annual lecture on research equity for international medical graduates in low- and middle-income countries), NeuroCamp (year-long neuroscience curriculum for Birmingham middle school students), and a spina bifida prevention campaign promoting corn masa folate fortification through community education.
- Chapter honors: 1st place, 2025 Global Hackathon (rural stroke care); Top 10, 2024 Global Hackathon; North America Service and Compassion Award.
- As Outreach Chair, direct fundraising, inventory, and equipment logistics supporting Sierra Leone's first publicly accessible neurosurgical service, in coordination with the WFNS and the Ministry of Health.
- Coordinate the roughly 35-member Global Neurosurgery Research Group; oversee a manuscript bundle mapping global neurosurgery stakeholders in line with the Boston Declaration, with additional manuscripts in preparation across the Global Stroke Outcomes Study (University of Utah), Sierra Leone, Hackathon, and Peer Mentorship committees.

**Biomedical Engineering Interest Group — Chair of Events**
Organized speaker events connecting medical students with UAB biomedical engineering faculty and neurotechnology research.

**Pulmonary Module — Course Representative**
Served as liaison between students and course faculty.

## Community Service and Advocacy

**Alabama House Bill 384 — Corn Masa Folic Acid Fortification**
Worked with the Children's of Alabama pediatric neurosurgery team to support passage of state legislation establishing folic acid fortification of corn masa, addressing the elevated prevalence of spina bifida in Hispanic communities.

**Medical Students Ask and Advise: Smoking Cessation — Co-Director**
Co-led bi-monthly smoking cessation sessions at The Foundry, a residential recovery program serving men at risk of drug-related re-offense.

**Students Helping At-Risk Patients (SHARP) — Program Member**
Work with a multidisciplinary team to help at-risk patients attain better access to healthcare.

**Lions Free Mobile Eye Clinic — Volunteer**
Performed vision screenings and intake for uninsured patients at free mobile eye clinics across rural Alabama.

**Talladega Superspeedway EMS — EMT Volunteer**
Provided EMS coverage as an EMT for race events at Talladega Superspeedway, delivering acute care for spectators and staff in a high-volume setting.

**Health Equity Scholar — Recipient and Volunteer**
Completed service-learning projects for each organ module, semester seminars, monthly social determinants of health grand rounds, and health equity research; on track for M.D. with Distinction in Health Equity.

## Work Experience

2024 — **GBR Lawncare LLC**
Interim CEO
- During the spring and summer of MS1 year, took over a family lawncare business during a parent's health issues, traveling two hours between hometown and Birmingham to mow and landscape 30 yards per week while completing pre-clinical coursework.

2020 – 2023 — **Family First Medicine, Rainsville, AL**
Medical Assistant
- Part-time medical assistant at a rural town's family practice.

2020 – 2023 — **Birmingham-Southern College Football**
Fullback, Tight End, Offensive Lineman
- Starting fullback for the BSC Panthers in fall 2021 and 2022; graduated in junior year to enter medical school a year early.

2014 – 2023 — **Hall Farms, Fyffe, AL**
Supervisor
- Mechanic and supervisor over farm and warehouse production lines for one of the largest potato farms in Alabama. Began as a manual laborer at 14 and worked up to supervisor through high school, overseeing 30–40 workers in picking, washing, sorting, bagging, and stacking hundreds of tons of red and Yukon gold potatoes.
- Oversaw all machine and tractor maintenance while planting, spraying, and harvesting over 1,000 acres of potatoes, cotton, pumpkins, and watermelons, including chemical treatment of all crops.